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Threat Intelligence Sharing in the Financial Sector

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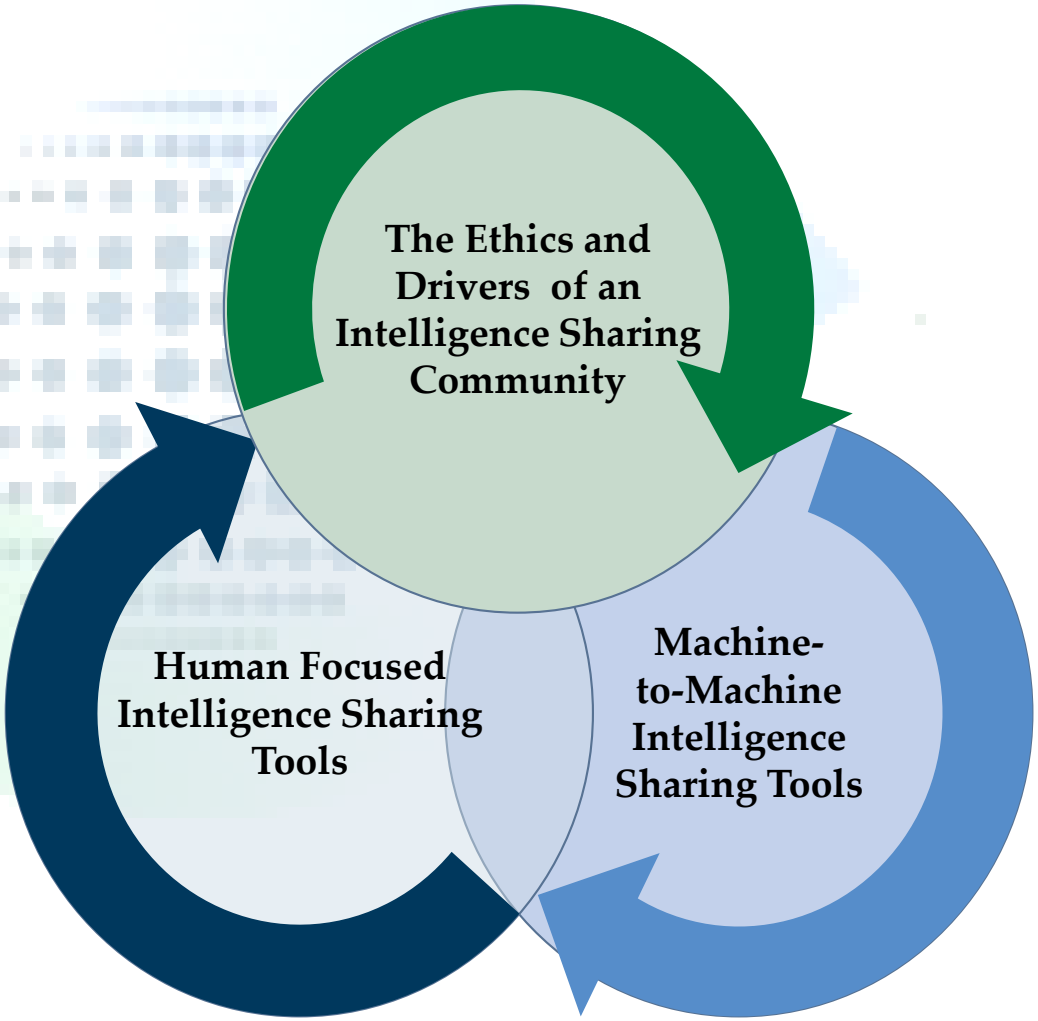
Financial Services
Information Sharing & Analysis Center

Sharing Communities Commonly Include Three Elements

Before any real sharing can begin a lot has to happen:

- **There has to be clear reasons for competitors to come together**
- **There have to be ways and means of coming together**
- **There have to be guidelines, rules or norms to define how the community operates**



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Today's Threats to Banking and Finance

39% of firms report being cybercrime victims in the last 12 months

41% of firms believe they will experience cybercrime within 24 months

13M Identity Fraud Victims in 2013

28% of fraud related to account takeovers

>€22B Expected IT security spending by 2017

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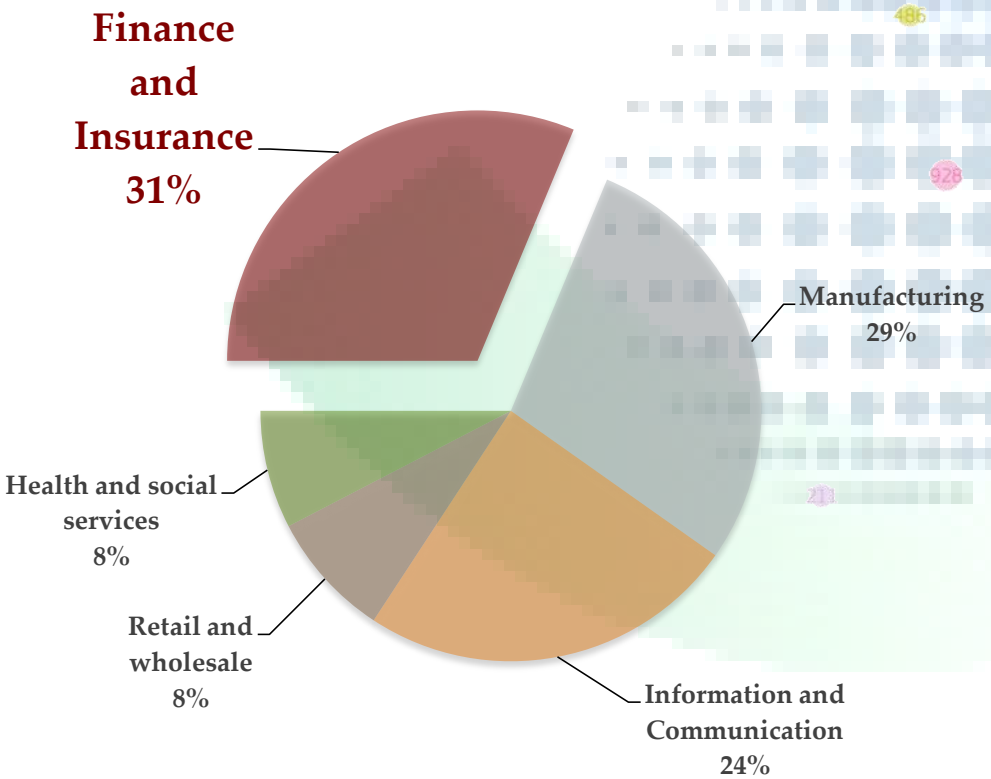
1) http://www.pwc.com/economiccrime
2) https://www.javelinstrategy.com/news/1467/92/A-New-Identity-Fraud-Victim-Every-Two-Seconds-in-2013-According-to-Latest-Javelin-Strategy-Research-Study/d,pressRoomDetail
3) http://www.canalys.com/newsroom/it-security-spend-reach-301-billion-2017

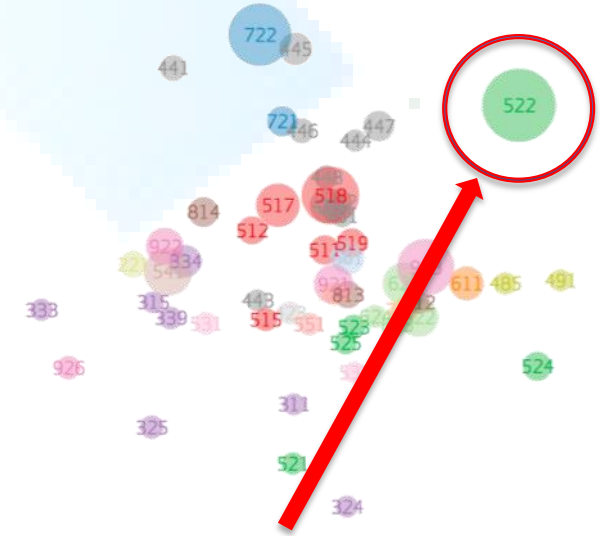
Target: Financial Sector

"Because That's Where The Money Is…" –Willie Sutton

Share of observed attacks by industry(2013)[1]



Uniqueness of breaches by industry (2014)[2]



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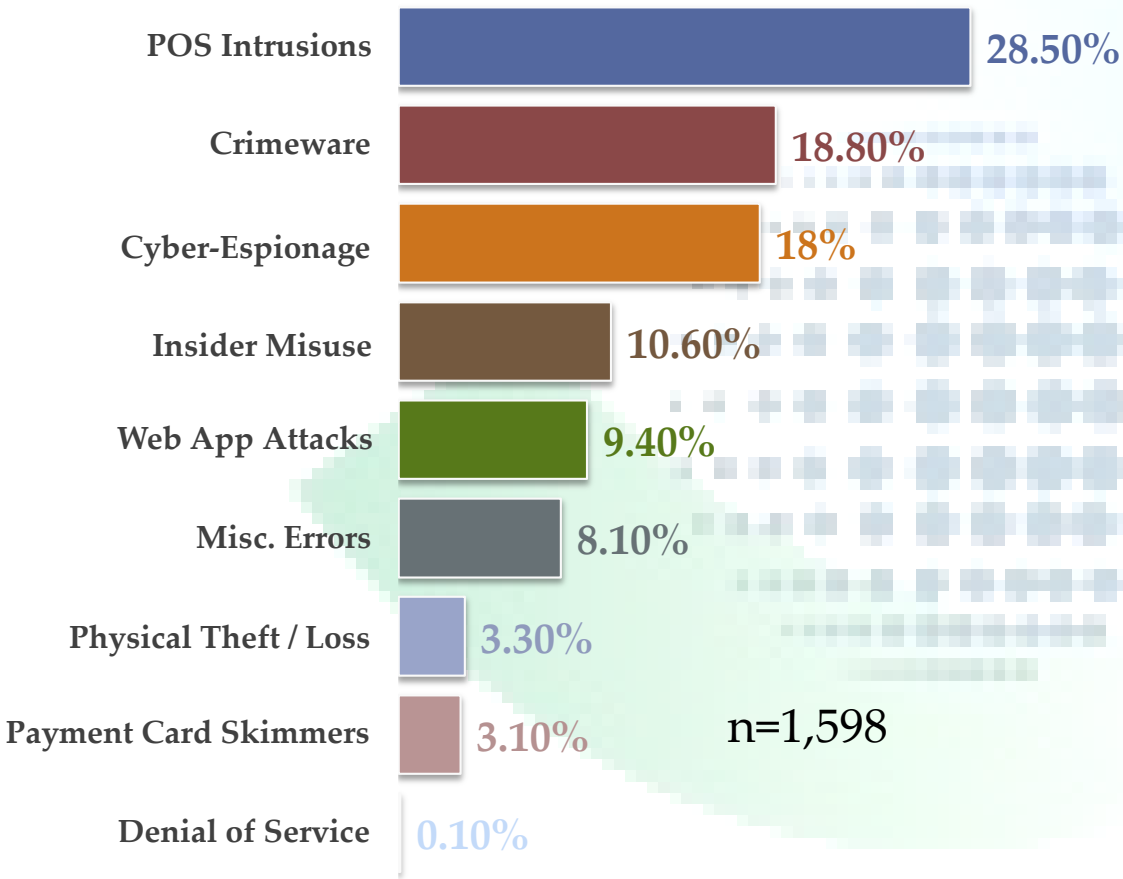
Sources: [1]*IBM 2014 Cyber Security Intelligence Index*
[2]*Verizon 2015 DBIR*

Frequency of Incident Classification Patterns

(Percentage Across Confirmed Data Breaches 2014)



- **Top concerns**: POS intrusions, web app attacks, espionage, and card skimmers
- **Threat profiles vary by region** (e.g.: European EMV less vulnerable to skimming, different mobile usage patterns, etc.)
- **Mobile does not appear to be a top issue**– but depends on your risk exposure
- **Phishing** is on the rise. Did it ever stop?

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Source: Verizon, [2015 Data Breach Investigations Report]

The Need For Speed

Attackers Act 150x Faster Than Victims Respond

	Seconds	Minutes	Hours	Days	Weeks	Months
Initial Attack to Initial Compromise (Shorter Time Worse)	10%	75%	12%	2%	0%	1%
Initial Compromise to Data Exfiltration (Shorter Time Worse)	8%	38%	14%	25%	8%	8%
Initial Compromise to Discovery (Longer Time Worse)	0%	0%	2%	13%	29%	54%

Attackers have honed their skills to come at you rapidly

Defenders take a long time to feel the impact of an attack

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Threats Evolve

CryptoLocker (2013)

CryptoWall (2014)
- *Version 1.0*
- *Version 2.0*
- *Version 3.0*

TorrentLocker (2014)

TeslaCrypt (2015)
- *Version 1.0*
- *Version 2.0*

CTB Locker (2015)

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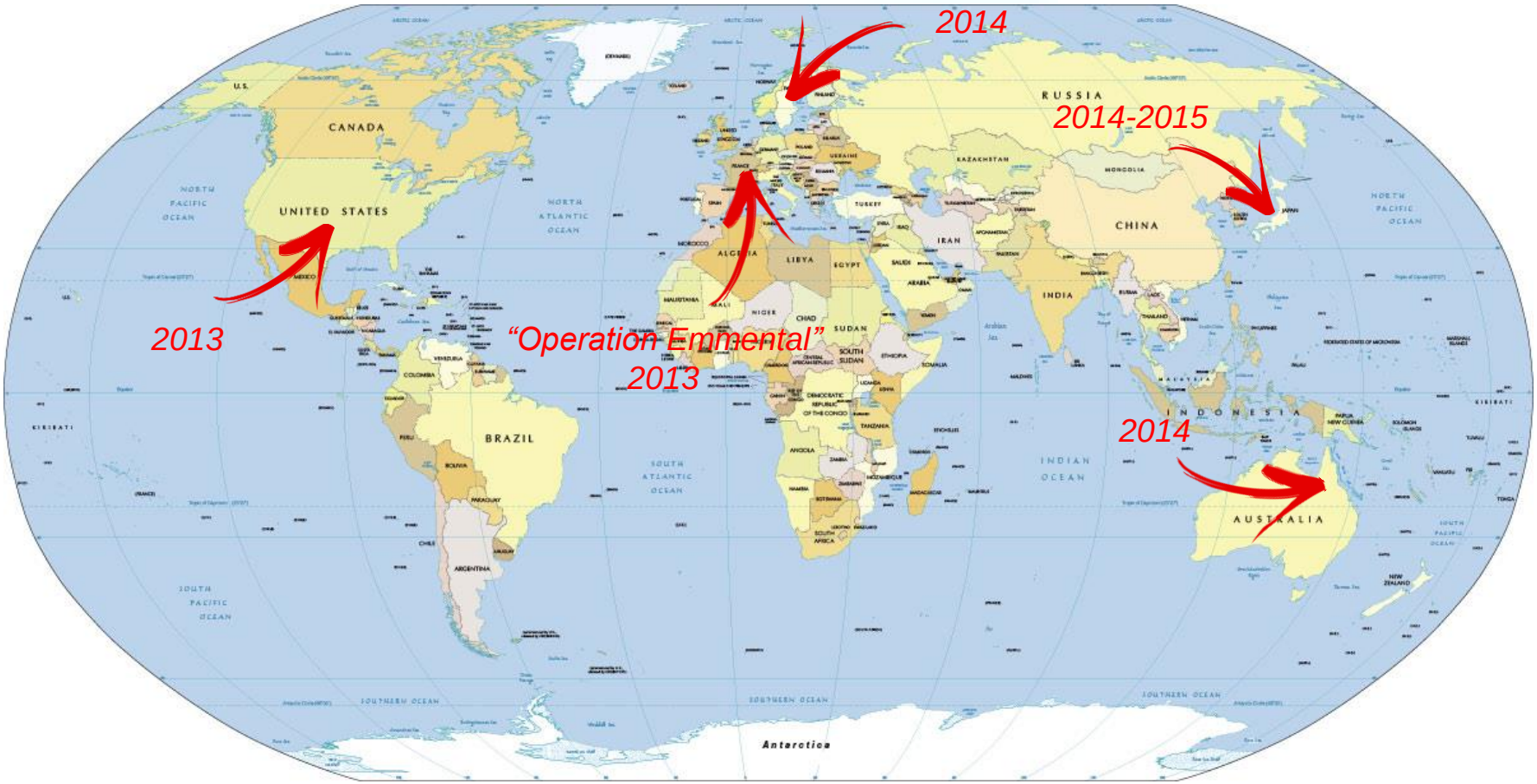
Threats Spread

E.g. Retefe Banking Trojan



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Attacks Become International

- Increasing sector- or region-specific threats and attacks
- Attacks spread across countries / regions
- Growing technological sophistication
- Ease of attacks

- Too much information
- Local / individual response
- Regulatory pressure

Cooperation Becomes Vital

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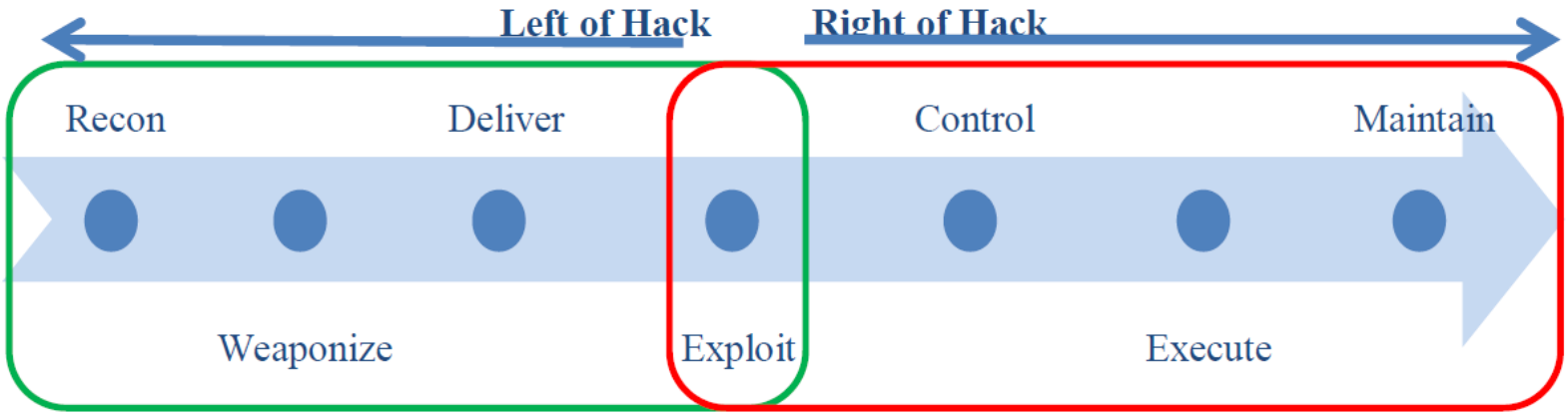
The Intel Sharing Business Case



Example Cyber Kill Chain

- Responding to incidents after the event is very costly.
- Proactive organisations need to move "left of the hack"
- It is not cost effective to do this with a purely compliance driven approach.
- This strategy requires **Cyber Threat Intelligence**.

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Source: Lockheed Martin [Intelligence-Driven Computer Network Defense]
Diagram: MITRE [Standardizing Cyber Threat Intelligence Information with STIX]

What is Threat Intelligence?

Tactics, Techniques, Procedures (TTP)

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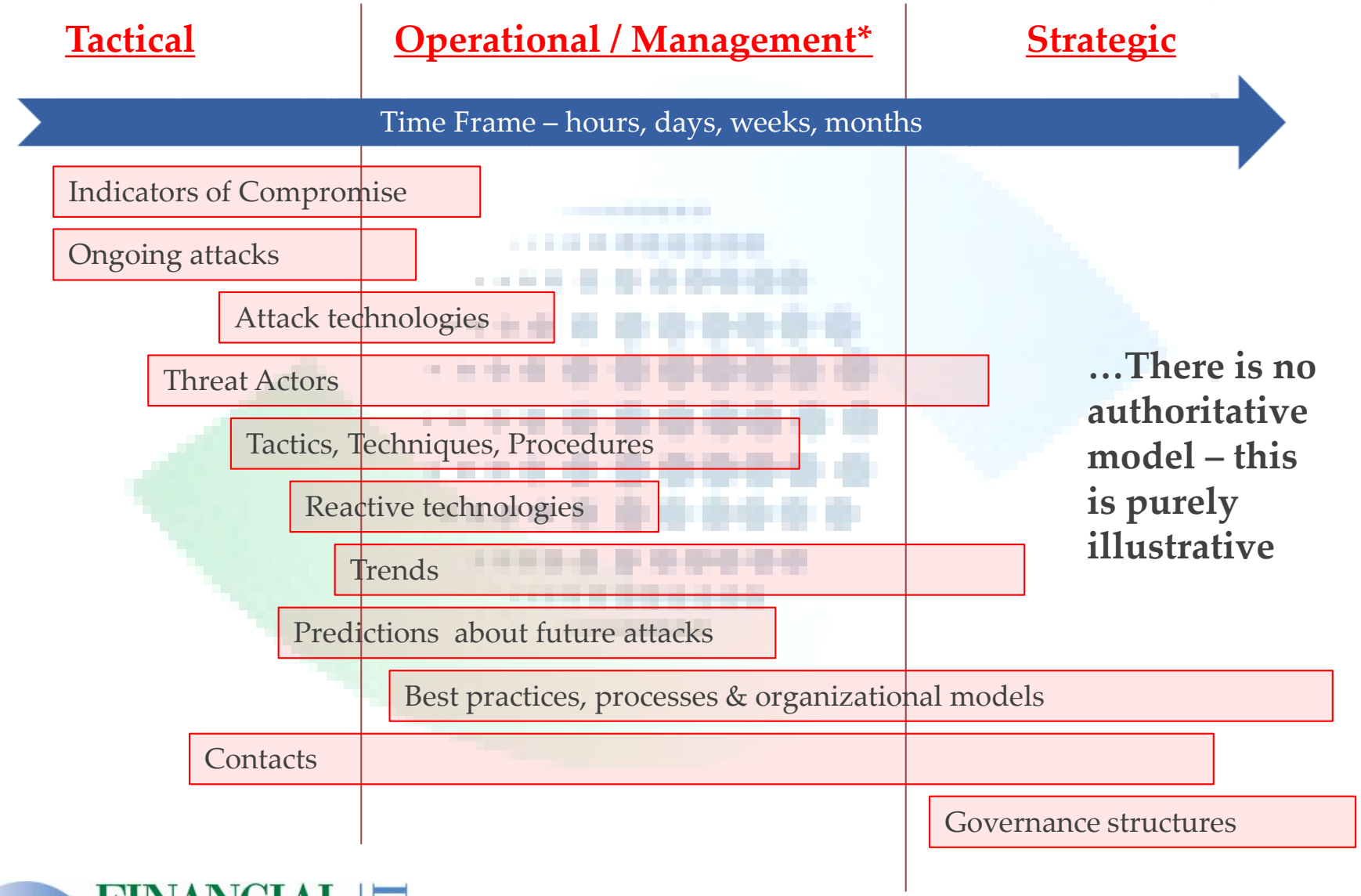
Levels of Information Sharing



…There is no authoritative model – this is purely illustrative

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Challenges to Intel Sharing in Europe (1)

The Will is Usually There…

- Diverse, Evolving Regulatory Requirements
  - Large FIs tend to rely on regulatory „stick“ to budget appropriately
  - ➢ Insecurity about what's coming – NIS directive, IT-Sicherheitsgesetz...what does it mean for us?
- Legal Blocks, Real or Perceived
  - Perceived legal blocks (e.g. data protection laws) to information sharing
  - Wide range of laws and perceptions of confidential data -> and what does the legal department say?
  - ➢ What may I share? What can't I share? With whom? → Numerous efforts in progress to define this
- Best Practices & Maturity Gaps
  - Intelligence sharing is still new for many organizations
  - Lack of visibility of incident management processes / intel analysis and sharing within firms
  - „Risk transparency“ is often a dirty, dirty phrase
  - ➢ How do I best benefit? What should processes look like?

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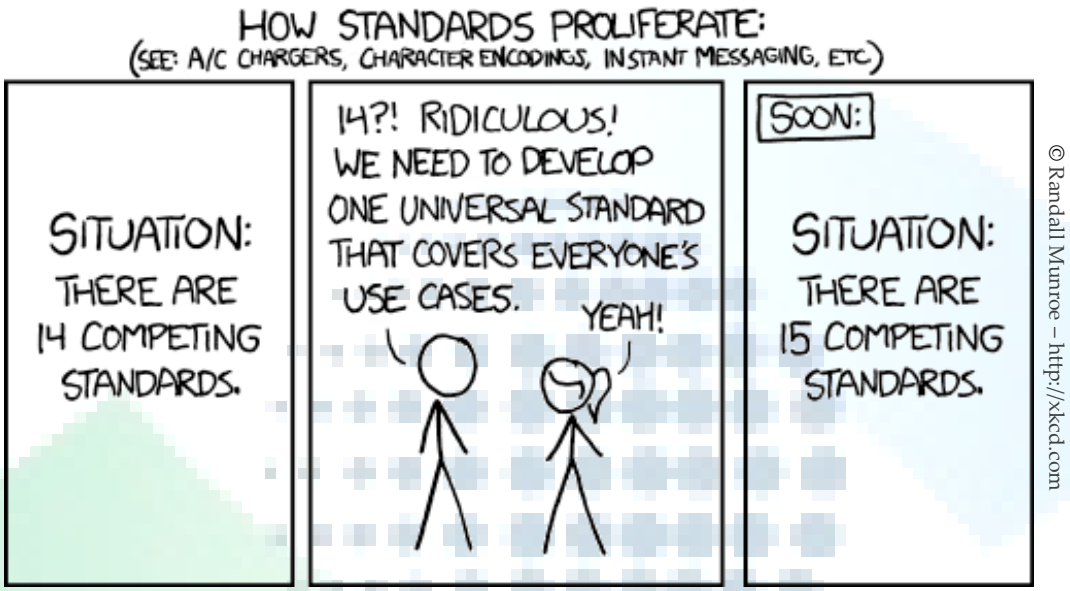
Challenges to Intel Sharing in Europe (2)

Who Is Responsible For What?



- Many arise in isolation – nature abhors a vacuum
- Some national, regional, sector capabilities very mature – what about international?
- Frequently the result of cultural, language factors
- Relationships often highly personal, ad hoc → fragmented landscape
- Cross-sector communication?
- ➢ Lack of clarity about responsibilities, maturity, points of contact, etc.

We should probably do something about these…

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Challenges to Intel Sharing in Europe (3)

No, Seriously, It Is.

- Too Much Information
  - Many organizations flooded with unfiltered intel
  - Much global intel not regionally / locally relevant
  - ➢ Helping members develop analysis and filtering capability, creating targeted intel flows
- Law enforcement / regulators?
  - Few standards for interacting with police and financial regulators in case of a breach
  - „The police are a black hole“
  - ➢ How to ensure incidents / threats / breaches are processed pragmatically? The right people are involved?
- Other Intel Sources: Competition vs. Complement?
  - „But we already have a provider“ -> Good! Keep them! Multiple sources of information are not mutually exclusive
  - Participating in an intel scheme helps leverage existing security expenditure
  - ➢ Bringing threat intelligence into a global context – not just geographically, but industry-wide

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A Few Points About „Information Sharing“

Everybody talks about it, but what does it mean?

- Just talking about Information sharing **is not** information sharing
  - ✓ Put up or shut up!
- The more you contribute, the more you get out of it
  - ✓ Participation motivates others to also participate
  - ✓ Sharing leverages and multiplies the value of existing feeds, contacts, and resources
- There is no „wrong“ way to share information
  - ✓ Threat intel, best practices, op models, processes, it‘s all useful to somebody
  - ✓ It‘s not just about your threat intel teams – also legal, investigations, etc.
  - ✓ Guess what? **Everyone has similar problems and questions as you do.**
- Threat intel is NOT A COMPETITIVE ADVANTAGE
  - ✓ Focus on your core business. Collaboration strengthens everybody
- Trust is everything
  - ✓ Whether rule-based or knowing your counterparts from face-to-face meetings

→ Participate in an information sharing scheme
→ Make sure you‘re talking to other schemes

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Imagine A World Where...

5000+ Financial services firms around the world actively sharing real-time threat information

12M Threat indicators automatically analyzed, prioritized, turned into actionable intelligence

5500 Prioritized incident & threat alerts shared/year

400 Key cyber intelligence email list posts / month

10x Reduction in time to respond to threat indicators via industry-created automation tools and feeds

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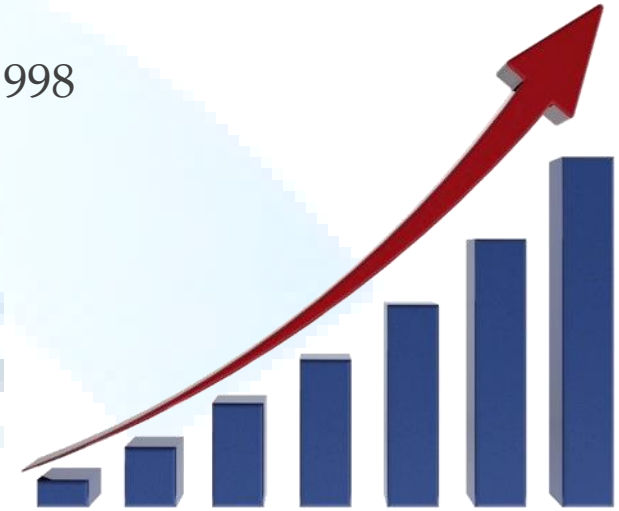
About FS-ISAC

MISSION: To Share Timely, Relevant, Actionable Digital and Physical Security Information & Analysis Specific to Financial Services



- A **not for profit** private sector initiative formed in 1998
- Created by financial services, for financial services
- Information sharing: both human and automated - to detect, prevent, respond to attacks & fraud
- Ca. 6,000 members on 6 continents
- Strong growth & traction globally in all major regions
- Regional / national staff: UK, continental Europe, US, Canada, Japan, Singapore; Security Operations Centers (SOC) in US, Poland

"The FS-ISAC is not a service provider, it's a community…Like neighborhood watch for electronic and physical hazards."

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FS-ISAC: Operational Intelligence Shared

Threats, Vulnerabilities, Incidents

- Malicious Sites
- Threat Actors, Objectives
- Threat Indicators
- TTPs, Observables
- Courses of Action
- Exploit Targets
- Denial of Service / DDoS
- Malicious Emails: Phishing/ Spearphishing
- Software Vulnerabilities
- Malicious Software (Malware)
- Analysis and risk mitigation
- Incident response techniques
- Attack trends & predictions

Via…

- ✓ Cyberintel mailing lists – US & EU, APAC, ES (Spanish-speaking)
  - 1500+ subscribers
  - 179 companies
  - 400+ posts/month
- ✓ Portal – with anonymous member submissions from 37 countries
- ✓ Biweekly threat calls – US, Europe, APAC
- ✓ Regular workshops, member meetings & briefings
- ✓ Threat Intelligence Committee (TIC)
  - 54 individuals / 26 companies

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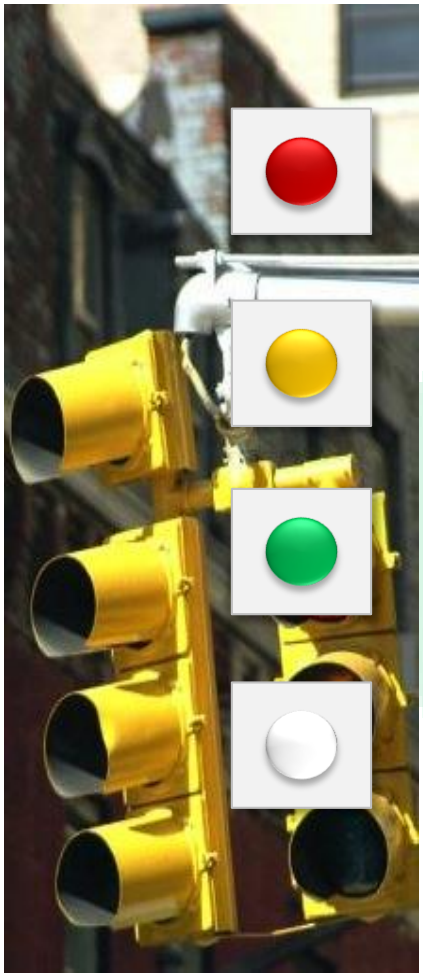
Information Sharing: Traffic Light Protocol



Ensuring the Appropriate Audience

- Restricted to a defined group (e.g., only those present in a meeting.) Information labeled RED should not be shared with anyone outside of the group
- This information may be shared with FS-ISAC members.
- Information may be shared with FS-ISAC members and partners (e.g., vendors, MSSPs, customers). Information in this category is not to be shared in public forums
- This information may be shared freely and is subject to standard copyright rules

Members must have full control over whom information is shared with.

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FS-ISAC Operations

Information

GOVERNMENT SOURCES
- Security Services
- FS Regulators
- Law Enforcement
- Other Intel Agencies

PRIVATE SOURCES
- iSIGHT Partners
- Fox-IT
- Wapack Labs
- NC4 PhySec Incidents
- Secunia Vulnerabilities

FS-ISAC 24x7 Security Operations Center

CROSS SECTOR SOURCES
- Cross Sector (Other ISACs)
- Open Sources (Hundreds)

Member Communications
- Information Security
- Physical Security
- Business Continuity / Disaster Response
- Fraud Investigations
- Payments / Risk

Member Reports Incident to Cyber Intel list, or via anonymous submission through portal → **Members respond in real time with initial analysis and recommendations** → **SOC completes analysis, anonymizes the source, generates alert to members. Police liaison <u>when appropriate.</u>**

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Member Submissions
Alerts

FS-ISAC's Working Groups

Circles of Trust Govern Member Interaction

- Clearing House and Exchange Forum (CHEF)
- Payments Risk Council (PRC) (EU PRC in development)
- Payments Processor Information Sharing Council (PPISC)
- European Threat & Strategy Committee (ETSC)*
- APAC Threat & Strategy Committee (ATSC)
- Business Resilience Committee (BRC)
- Threat Intelligence Committee (TIC)
- European Threat & Strategy Committee (ETSC)
- Community Institution Council (CIC)
- Insider Threat Working Group (ITWG)

- Insurance Risk Council (IRC)
- Compliance and Audit Council (CAC)
- Cyber Intelligence Mailing Lists (Currently Intl., EU, APAC, ES)
- Education Committee
- Product and Services Review Committee
- Survey Review Committee
- Security Automation Working Group (SAWG)
- JWGI - European Regulatory & Legal Overview Working Group*
- JWGI - European Threat Intel Sharing Landscape Working Group*
- EU CAPP development group*

...Plus strategic cooperation with outside groups – banking federations, law enforcement, intel sharing schemes, CERTs, WGs, etc.

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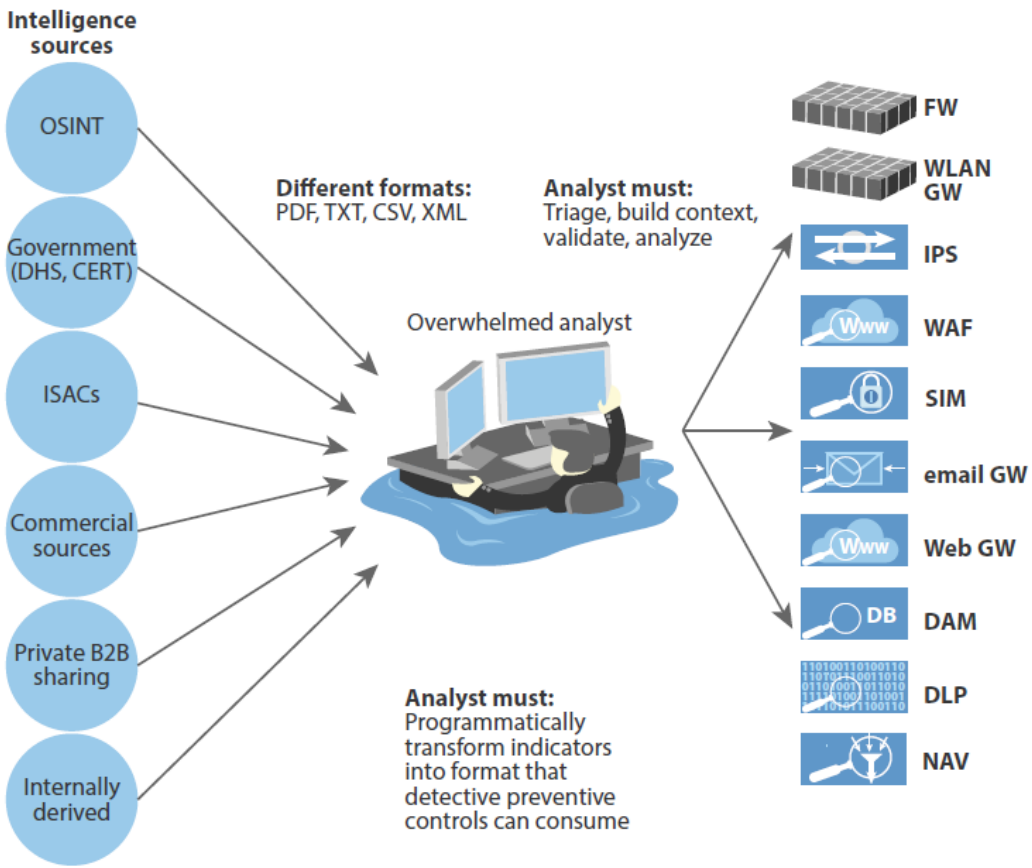
Europe-specific groups. Other groups are inter-regional

Intelligence Data Overflow

Figure 2 Acting On Intelligence Overwhelms Analysts



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Source: Forrester Research, Inc.

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Threat Intelligence Standards

The Power of Structured Intelligence

- Key to effective strategic cyber intelligence analysis and threat tracking
- Ability to pivot, view, analyze, and enrich complex relationships
- Threat Language that Machines Speak ◎ STIX
  - Verbose
  - Structured
- Common Transport ◎ TAXII
  - Restful
  - Optimized for STIX
- Open standard supported by OASIS.

STIX

Structured Threat Intelligence eXpression

TAXII

Trusted Automated eXchange of Indicator Information

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www.oasis-open.org/committees/cti

STIX Constructs

An open standard to categorize cyber threat intelligence information

Atomic

What threat activity are we seeing?

Tactical

What threats should I look for on my networks and systems and why?

Operational

Where has this threat been seen?

What can I do about it?

What weaknesses does this threat exploit?

Strategic

Who is responsible for this threat?

Why do they do this?

What do they do?

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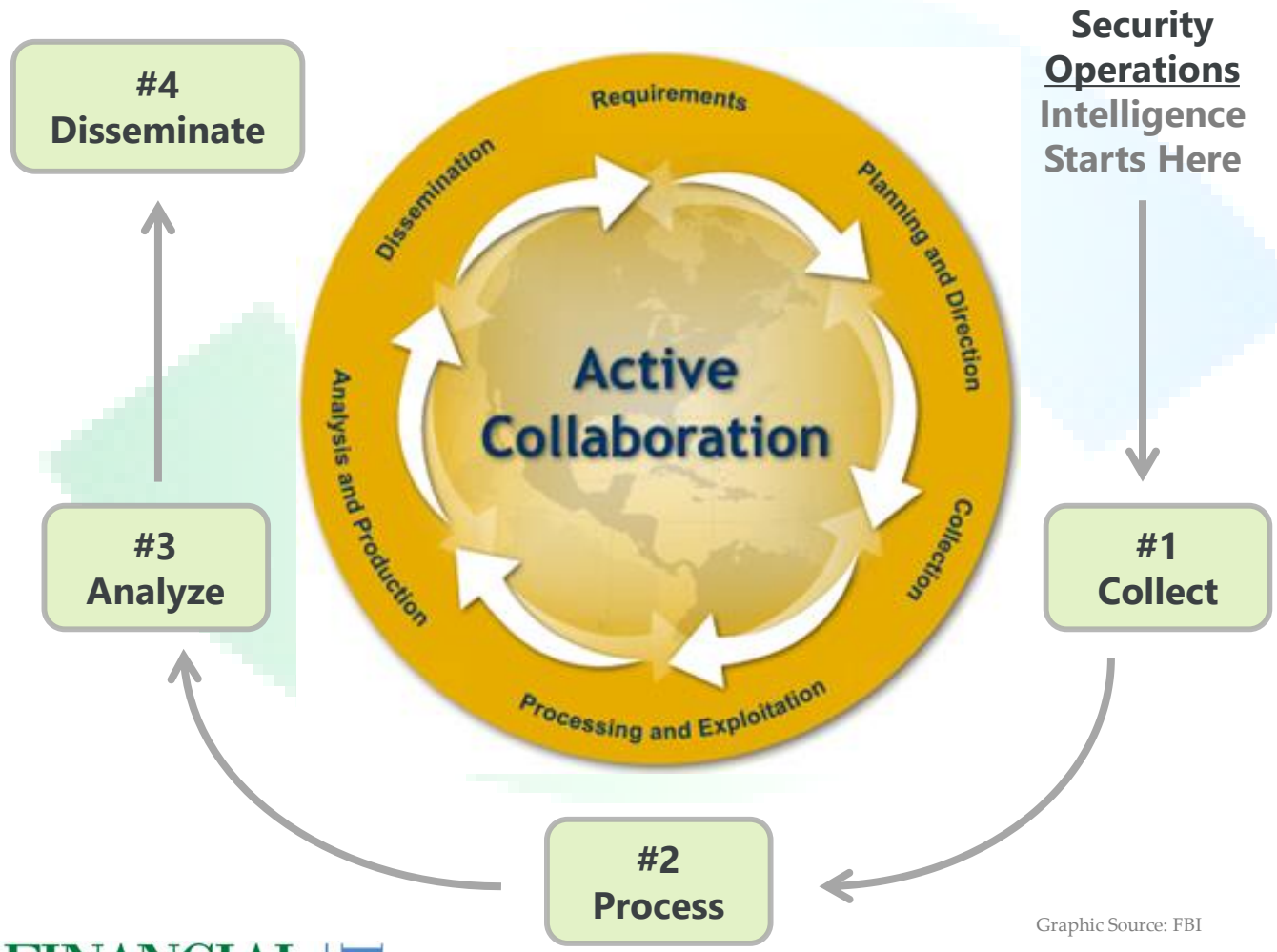
The Intelligence Lifecycle

What Do We Do With It? **(What are we supposed to do with it?)**



Graphic Source: FBI

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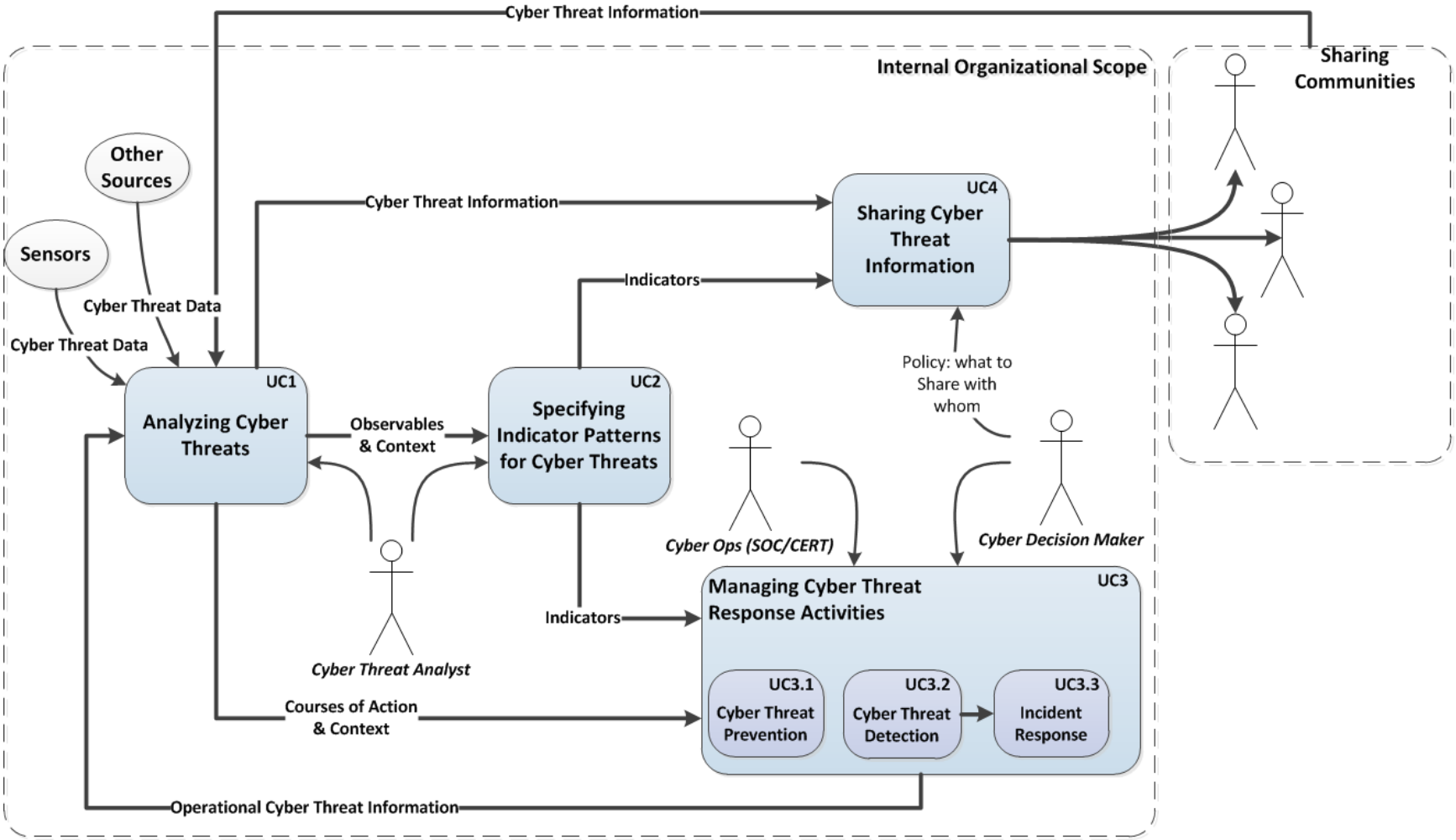
STIX Supported Use Cases



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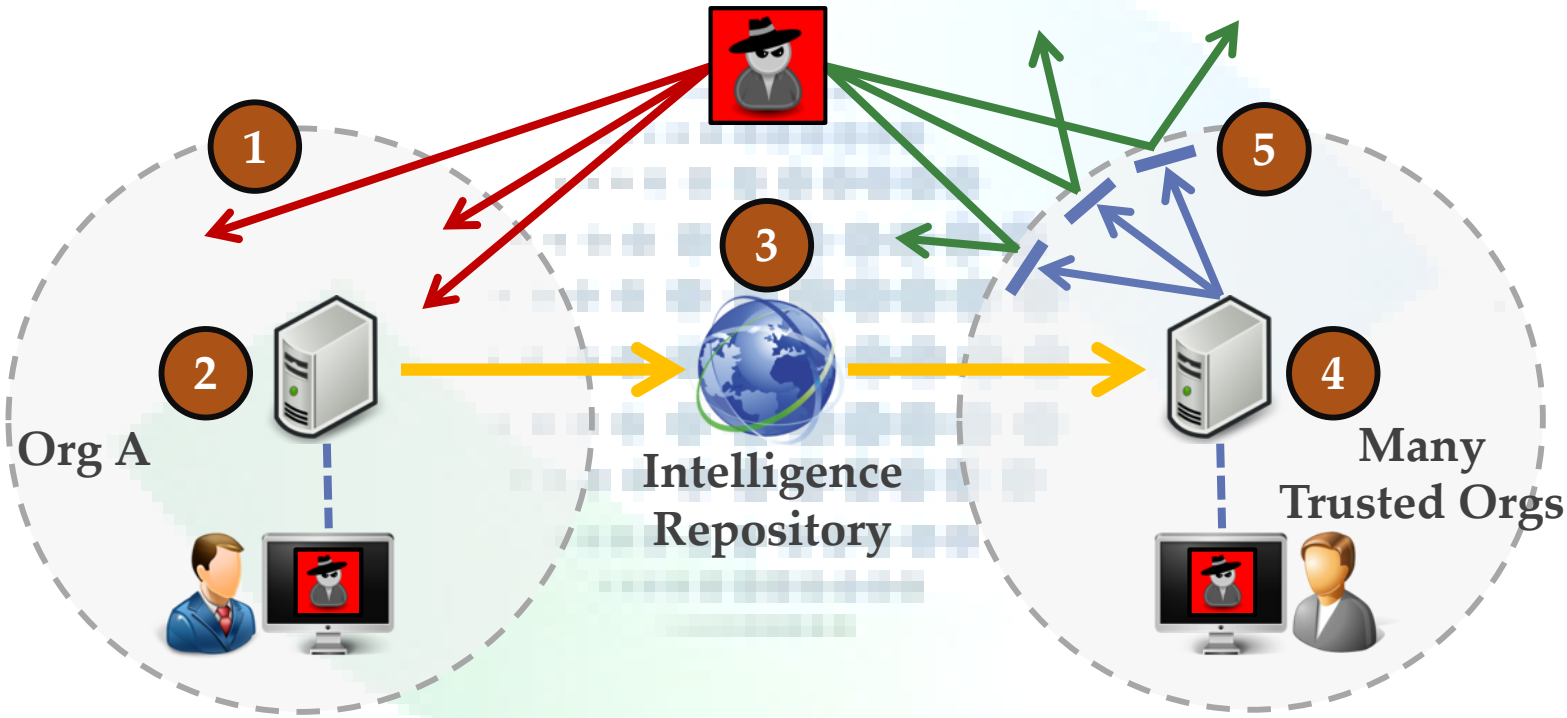
Threat Intelligence Automation

Instead of 2% or less of attacks blocked, detected, or prevented…



…a much higher percentage of attacks are stopped.

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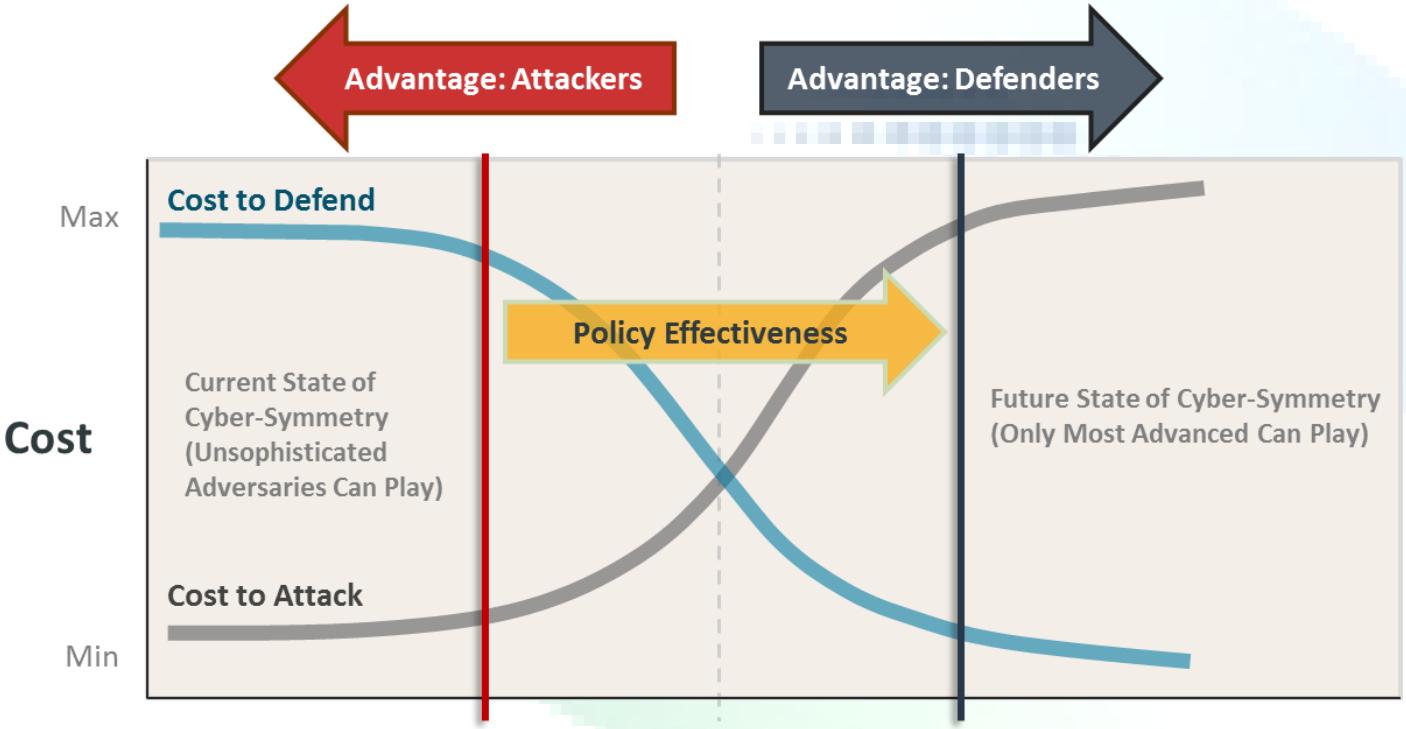
Cyber Threat Economics 101



Cost to Firms ↓
The current cost to process a single piece of intelligence is 7 hours

Cost to Adversaries ↑
Adversaries must "re-tool" much more often and their exploits cause less damage

Cyber Risks ↓
Frequency and impact of threats decrease, higher adoption => exponential benefits

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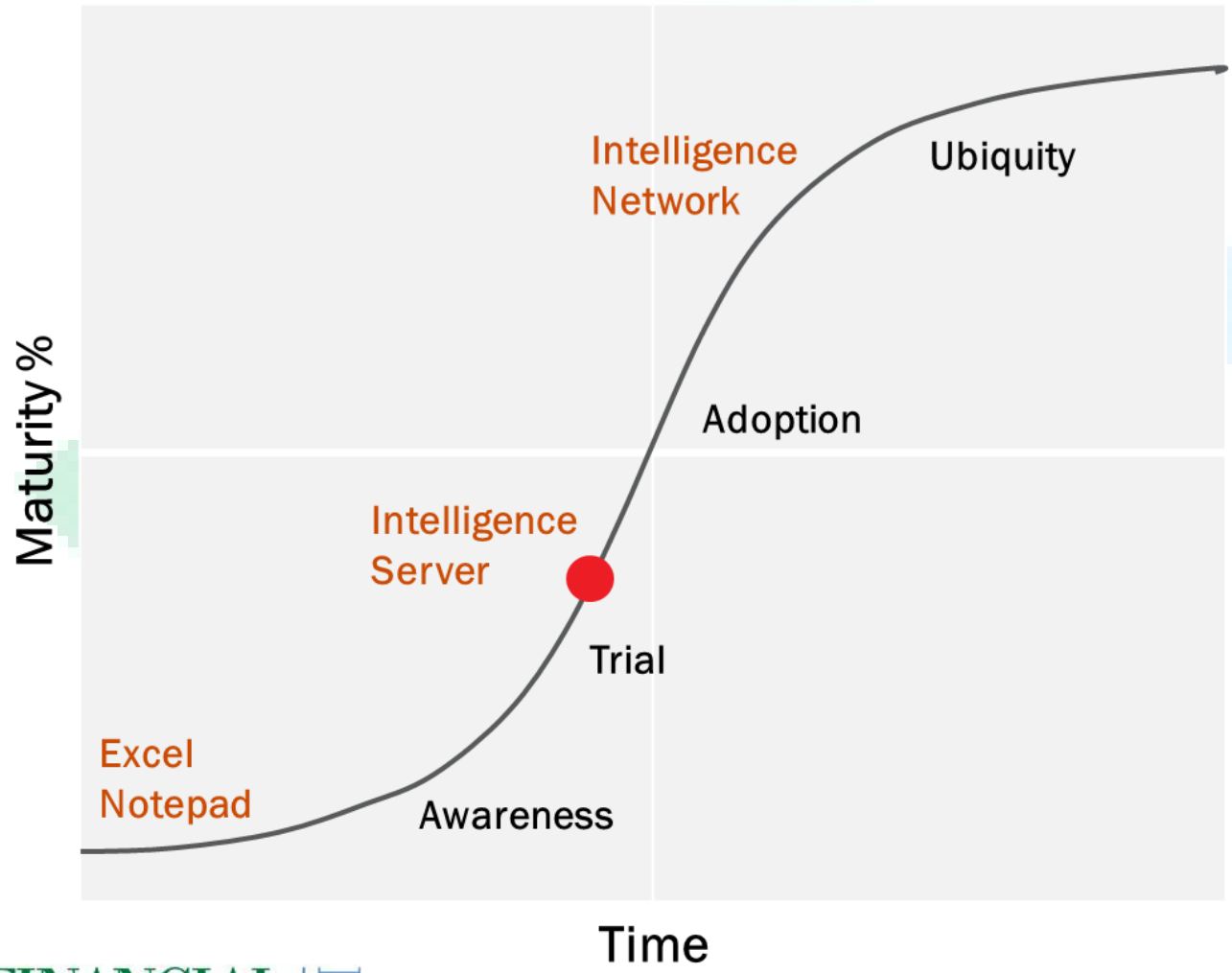
Standards Adoption Today



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Soltra Edge

An FS-ISAC and DTCC Company

SOLTRA™

- Soltra is a joint venture between FS-ISAC and DTCC (Despository Trust and Clearing Corporation).
- Soltra Edge is a threat intelligence server that can aggregate and distribute threat content within an organization and between organizations
- Free to use, support available, download from www.soltra.com
- Soltra Net is coming in Q4 and will simplify connection of Soltra instances and include strong authentication.

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Your Next Steps With FS-ISAC…

Prospective Members: Try Our Services

- ✓ Try FS-ISAC for 6 months at no cost
- ✓ Access the threat indicators
- ✓ Get information from the FS-ISAC portal
- ✓ Participate fully in FS-ISAC events, summits, simulations…and help us shape them locally, regionally and globally to meet your needs
- ✓ Find out why 6,000 other institutions are members!

Existing Members: Leverage all that FS-ISAC Offers!

- ✓ Member portal: Educate more users on access to portal, threat indicators, playbooks, content, security automation
- ✓ Education: Executive briefing to update others on FS-ISAC
- ✓ Participation: Attend Summits, regional events, threat briefings, threat simulations and webinars

→ **If there is anything we can do to help, please let us know!**

"This year's threat simulation with FS-ISAC was a terrific exercise and very helpful for our executive management team in understanding the true, complex nature of today's threat environment…"

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Contact Details

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Thank You For Your Interest in FS-ISAC

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